

**SPATIOTEMPORAL DISPERSAL PATTERNS OF THE MAYFLY *EPHEMERA*
DANICA (MÜLLER, 1764) ARE RELATED TO BODY MASS AND SIZE**

Running title: Dispersal of the mayfly *Ephemera danica*

**Leon A.H. van Kouwen^{✉,1,2}, Ralf C.M. Verdonschot³, Michiel H.S. Kraak¹, Piet F.M.
Verdonschot^{1,3}**

[✉] Corresponding author: l.vankouwen@has.nl

¹Institute for Biodiversity and Ecosystem Dynamics, Department of Freshwater and Marine
Ecology, University of Amsterdam, Amsterdam 1098 XH, Netherlands

²HAS green academy, 's-Hertogenbosch 5223 DE, Netherlands

³Wageningen Environmental Research, Wageningen University and Research, Wageningen 6708
PB, Netherlands

ORCID iDs: L.A.H. van Kouwen, <https://orcid.org/0000-0003-2918-5048>; R.C.M. Verdonschot,
<https://orcid.org/0000-0002-0977-5975>; M.H.S. Kraak, not available; P.F.M. Verdonschot,
<https://orcid.org/0000-0002-4126-7452>

SUPPLEMENTARY DATA

SUPPLEMENTARY DATA A – NYMPHS AND ENVIRONMENTAL CONDITIONS

The density of nymphs (Table 1) was determined by counting the nymphs in three samples per transect taken with a surber sampler of (0.12 m² in area) on 10/05/2024, totaling 12 samples per reach. Substratum (in-stream) was determined in September and October 2024, using the “Gewässerstrukturgütekartierung in der Bundesrepublik Deutschland” (Zumbroich and Müller, 1999) at each transect, thus totaling five observations per reach. Vegetation structure was determined at each location where LED traps were positioned (five transects with five LED traps, thus 25 plots per reach in total), using the Braun-Blanquet method (Westhoff and van der Maarel, 1973), in September and October 2024. Shading was analyzed using the R-package “CoveR2” (Alivernini et al., 2018) on photos taken on 10/05/2024.

Table 1 Nymphs, and environmental conditions in the upstream reach and downstream reach, with mean values and standard deviation (SD). Parameters: density of the nymphs, in-stream substratum (CPOM = Course Particulate Organic Matter, FPOM = Fine Particulate Organic Matter), and shading and vegetation in the riparian zone

position	parameter	Unit	Upstream reach		Downstream reach	
			Mean	SD	Mean	SD
Nymphs		Ind. / m ²	150	121	292	294
Substratum (in-stream)	Wood	%	2	2	17	6
	CPOM	%	5	2	5	2
	FPOM	%	9	6	10	4
	Course gravel	%	0	0	0	2
	Fine gravel	%	1	1	0	3
	Sand	%	85	8	65	7
	Silt	%	0	0	0	0
Vegetation structure and shading (riparian zone, 0-150m)	Shading	%	96	6	89	10
	Herbs	%	20	24	1	3
	Moss	%	0	4	2	19
	Shrubs	%	0	10	0	0
	Trees	%	25	10	15	11

SUPPLEMENTARY DATA B – WEATHER CONDITIONS DURING THE SAMPLING PERIOD

Table 2 Nightly weather data (20:00-04:00), with the date referring to the preceding night. Parameters: number of females and males, the percentage of females, water temperature (mean, minimum, and maximum), water level above the stream bottom (mean, minimum, and maximum), wind speed (mean, minimum, maximum) in both km/h and Beaufort (Bft) scale, and precipitation

Parameter		12/05/2024	13/05/2024	14/05/2024	18/05/2024	19/05/2024	22/05/2024
Number of individuals (N)	Females	42	23	191	6	33	16
	Males	10	12	28	0	7	1
Females (% of the total)		81	66	87	100	82	94
Water temperature (°C)	Mean	15.8	16.3	16.3	14.9	15.3	14.7
	Min.	14.5	14.8	15.4	13.4	14.3	14.1
	Max.	17.1	18.3	17.5	16.7	16.4	15.6
Water level (cm)	Mean	26.0	25.9	25.1	30.6	27.6	40.4
	Min.	25.9	25.2	24.9	27.5	26.6	35.4
	Max.	26.5	26.3	25.4	31.9	29.3	47.3
Air temperature (°C)	Mean	16.3	16.0	14.7	12.9	11.9	11.5
	Min.	13.2	12.6	11.0	11.9	11.0	8.9
	Max.	19.0	18.4	17.7	14.0	12.8	13.9
Wind speed (km/h)	Mean	3	2	2	2	2	2
	Min.	1	1	1	1	1	1
	Max.	3	3	3	3	2	3
Wind speed (Bft)	Mean	2	2	2	1	1	2
	Min.	1	1	1	1	1	1
	Max.	2	2	2	2	2	2
Precipitation (mm)		0	0	0	0	0	0

40 **Table 3** Daily weather data (00:00-23:59). Parameters: water temperature (mean, minimum, and maximum), water level above the stream bottom (mean,
41 minimum, and maximum), wind speed (mean, minimum, maximum) in both km/h and Beaufort (Bft) scale, and precipitation, with sampling days shaded grey.

Parameter		09/05/2024	10/05/2024	11/05/2024	12/05/2024	13/05/2024	14/05/2024	15/05/2024	16/05/2024	17/05/2024	18/05/2024	19/05/2024	20/05/2024	21/05/2024	22/05/2024	23/05/2024	24/05/2024	25/05/2024
Water temperature (°C)	Mean	12.6	13.1	14.1	14.9	15.8	16.3	16.3	15.5	15.0	14.9	14.9	15.3	15.3	15.4	14.7	13.7	14.4
	Min.	11.0	11.4	12.6	13.3	14.5	14.8	15.4	14.7	14.3	14.1	13.4	14.3	13.9	14.9	14.1	13.0	13.8
	Max.	14.1	15.0	16.0	17.0	17.1	18.3	17.5	16.2	15.6	15.7	16.7	16.4	16.8	16.1	15.6	14.6	15.1
Water level (cm)	Mean	30.4	29.2	27.8	26.6	26.0	25.9	25.1	25.7	26.0	25.7	30.6	27.6	26.1	43.0	40.4	33.5	38.6
	Min.	29.6	28.5	27.0	26.0	25.9	25.2	24.9	24.8	25.8	25.2	27.5	26.6	25.2	28.8	35.4	32.3	33.4
	Max.	31.1	29.5	28.4	27.0	26.5	26.3	25.4	26.3	26.2	26.8	31.9	29.3	28.4	51.6	47.3	35.4	41.7
Air temperature (°C)	Mean	12.8	15.1	16.4	19.4	19.5	21.2	17.0	15.9	16.0	16.2	15.4	15.5	17.2	15.2	14.7	14.0	15.0
	Min.	4.5	5.4	10.1	12.5	13.2	14.8	11.0	12.1	11.5	10.0	11.4	11.0	11.8	12.3	10.2	8.8	10.3
	Max.	19.4	21.2	22.9	25.1	24.5	27.4	23.0	21.3	20.1	22.0	22.0	21.4	23.8	18.7	19.7	17.4	18.3
Wind speed (km/h)	Mean	2	2	3	4	3	5	2	2	2	2	2	2	3	4	3	2	3
	Min.	0	0	2	2	1	2	1	1	1	1	1	0	1	1	0	0	1
	Max.	3	4	4	5	5	8	5	5	4	5	5	3	5	5	6	4	4
Wind speed (Bft)	Mean	0	0	2	2	1	2	1	1	1	1	1	0	1	1	0	0	1
	Min.	0	0	2	2	1	2	1	1	1	1	1	0	1	1	0	0	1
	Max.	0	0	2	2	1	2	1	1	1	1	1	0	1	1	0	0	1
Precipitation (mm)		0	0	0	0	0	0.3	0	2.4	2.4	2.5	0.7	20.7	0	25.4	2.1	0	7.7

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SUPPLEMENTARY DATA C – OVERVIEW, DISTRIBUTION AND CORRELATION
ANALYSES OF TRAITS

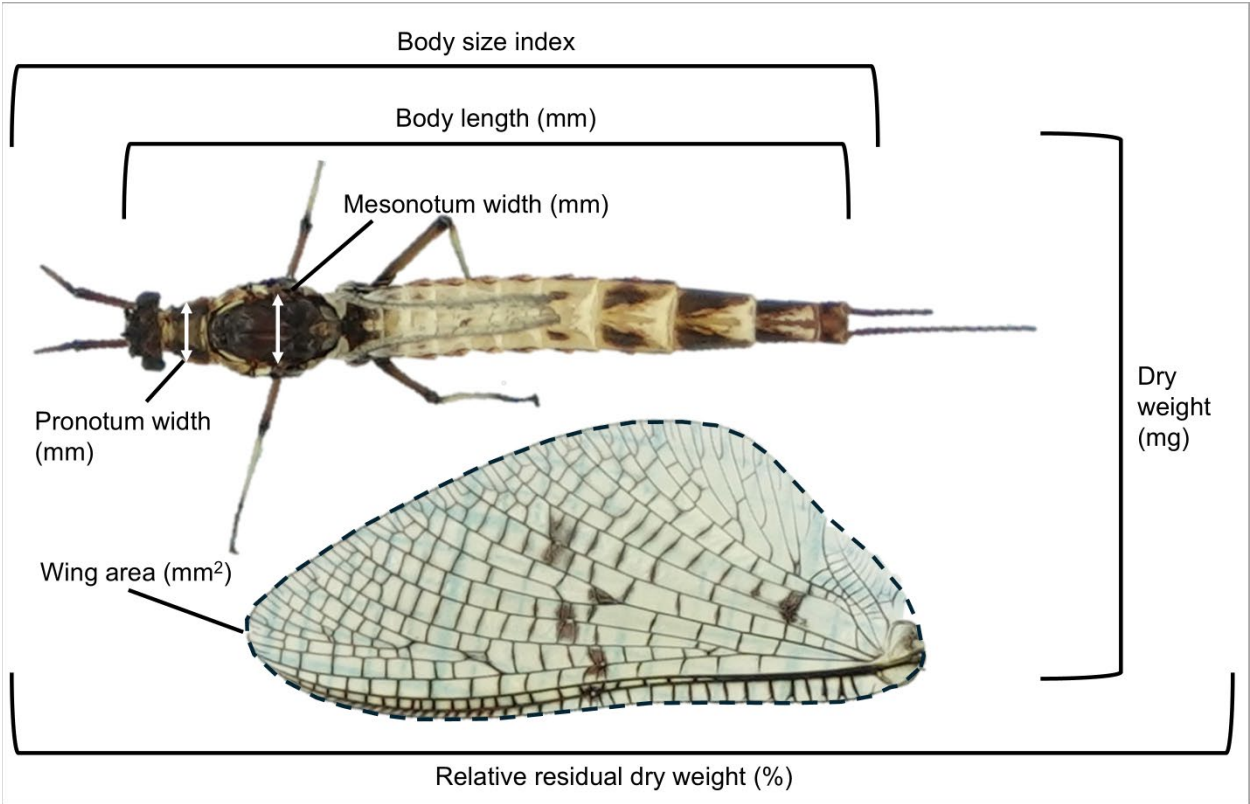


Fig. 1 Overview of the body size and weight related traits of the adult *E. danica* individuals, including the body size index, body length, pronotum width, mesonotum width, wing area, dry weight and relative residual dry weight, expressed as the percentage of dry weight not explained by body size, specifically eggs and/or lipids.

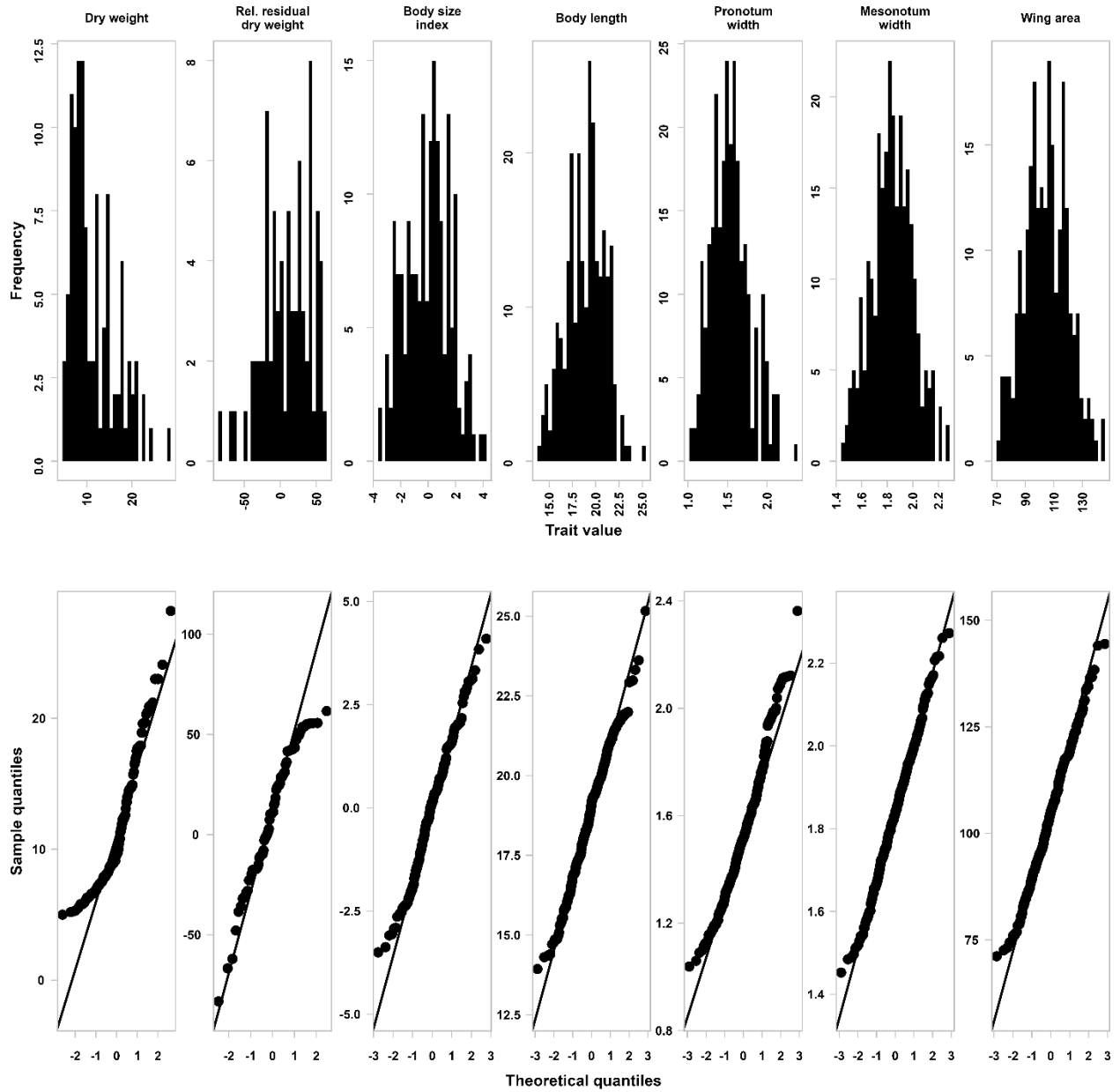


Fig. 2 Histograms (upper panel) and Q-Q plots (lower panel) of the measured traits for female *Ephemera danica* adult individuals

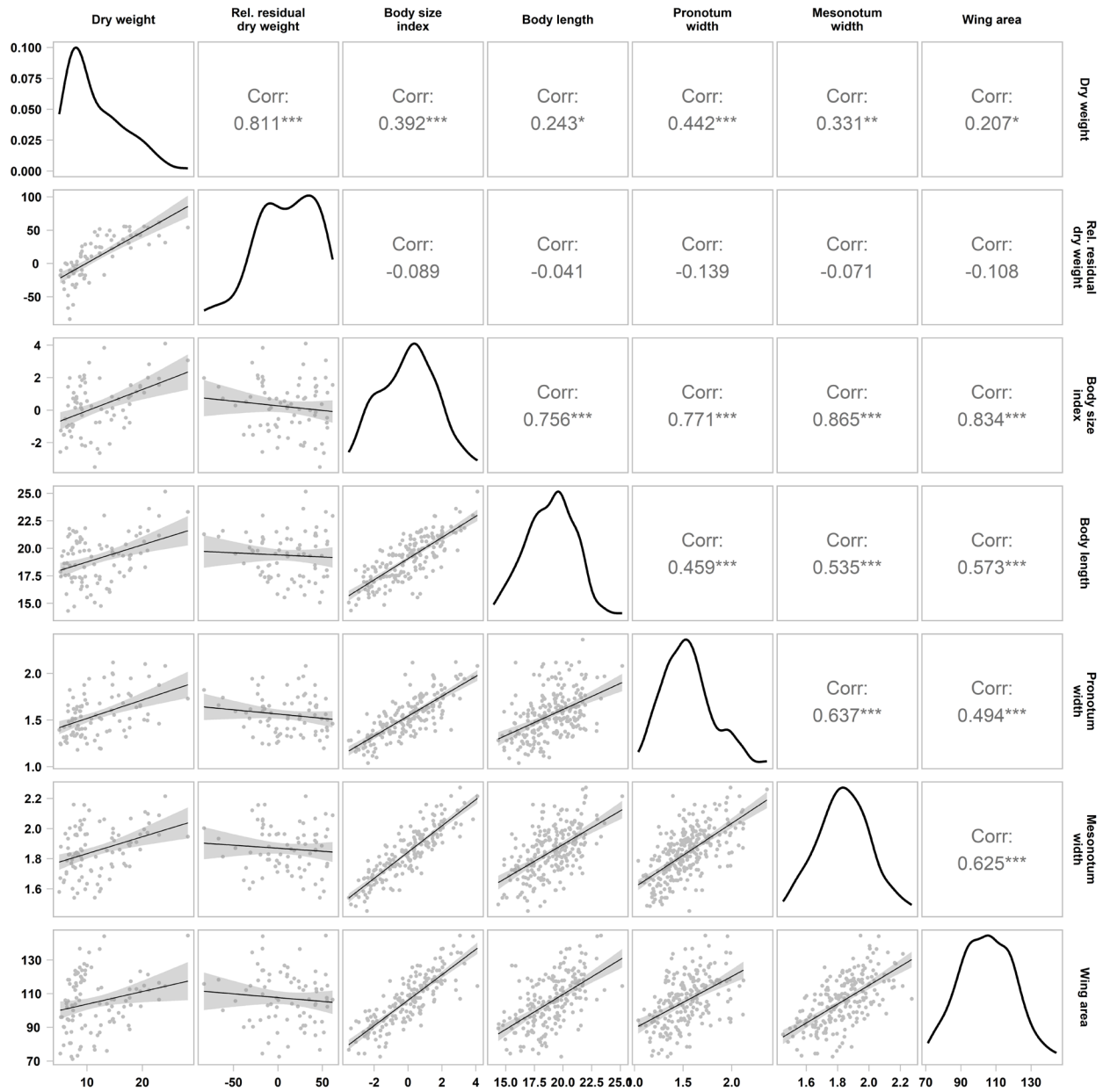


Fig. 3 Correlation of all female *Ephemera danica* traits. Bottom panels represent scatterplots with linear regression lines, middle panels display density plots of the respective parameters, and upper panels provide correlation coefficients (Corr, R). The plots were generated using GGally (Schloerke et al., 2010)

SUPPLEMENTARY DATA D – PCA CHARACTERISTICS

Table 4. PCA statistics per PCA axis, including standard deviation, Proportion of variance, and cumulative proportion. PC1 was identified as the principal component related to body morphology.

Statistic	PC1	PC2	PC3	PC4
Standard deviation	1.62	0.77	0.67	0.58
Proportion of variance	0.66	0.15	0.11	0.08
Cumulative proportion	0.66	0.80	0.92	1.00

Table 5. Loadings of the original variables on the principal components, indicating the contribution and direction of each variable to the PCA. PC1 was identified as the principal component related to body morphology.

Parameter	PC1	PC2	PC3	PC4
Body length	0.477	-0.627	0.616	-0.012
Wing area	0.511	-0.274	-0.665	0.472
Pronotum width	0.478	0.704	0.354	0.390
Mesonotum width	0.533	0.193	-0.232	-0.791

SUPPLEMENTARY DATA E – FREQUENCY PLOTS OF TRAITS FOR FEMALES AND
 MALES

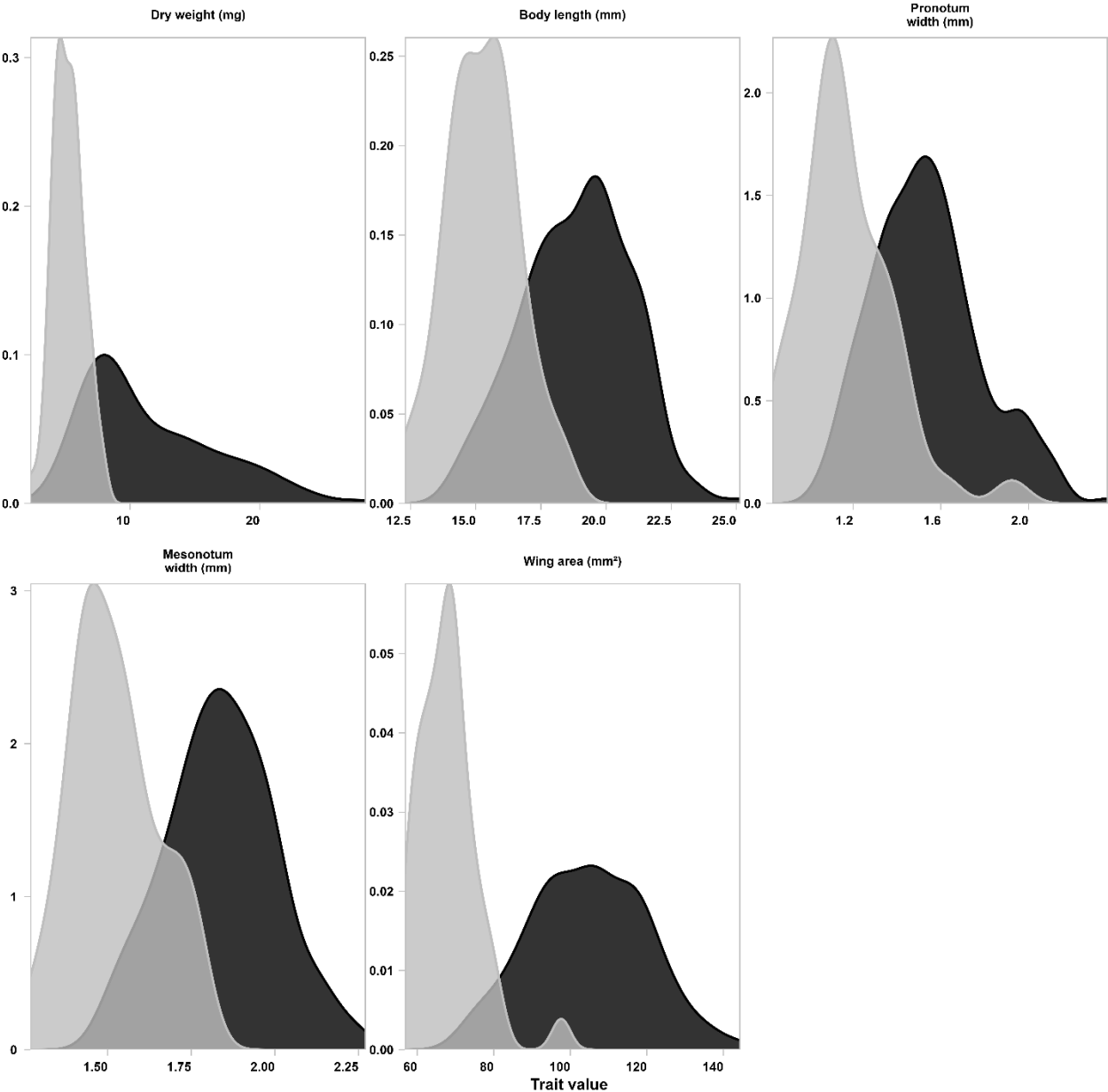


Fig. 4 Frequency plots of traits female (black areas), and male (grey areas) *Ephemera danica* adults

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